

Betsy -- DOT Framework Method Usage

Maps every DOT research strategy used across the project to the DL it appears in and the evidence files that support it.

Summary table

DL	Library	Showroom	Lab	Field	Workshop
DL-01			?		?
DL-02	?	?	?		
DL-03	?	?			
DL-04			?	?	
DL-05			?		
DL-06			?		
DL-07			?		
DL-08			?		

By strategy

Library

Desk research into existing tools, frameworks, or literature.

DL-02 -- Agent framework selection Compared LangGraph, AutoGen, CrewAI, and n8n against Betsy's procurement requirements before choosing LangGraph. Evidence:

PIPELINE_ARCHITECTURE.md · ORCHESTRA_ARCHITECTURE.md

DL-03 -- UI pattern research Studied how Salesforce Einstein, Microsoft Copilot, Tableau Pulse, Glean, and Notion AI present AI to non-technical business users. Extracted four recurring patterns (command bar, narrative, AI-scored tables, inline approval cards).

Evidence: docs/user_requirements.md · docs/wireframes.md

Showroom

Building or demonstrating prototypes to compare approaches or validate a direction.

DL-02 -- Pipeline vs Orchestra comparison Both architectures built and run against the same 4 scenarios. Outputs compared directly to identify the failure case (sequential processing of simultaneous stockout + duplicate invoice). Evidence: graph.py · decide.py ·

inventory_monitor.py · supplier_scout.py · invoice_auditor.py ·

PIPELINE_ARCHITECTURE.md · ORCHESTRA_ARCHITECTURE.md

DL-03 -- Interactive wireframe prototype dashboard/wireframe.html built as a fully interactive HTML prototype with working navigation, approve/decline buttons, expandable log entries, and a live badge counter -- used to validate layout and interaction model before

touching the real dashboard. Evidence: dashboard/wireframe.html ·
images/dl03-wireframe-comparison.png · images/dl03-wireframe-interactive.png

Lab

Controlled technical experiments: building, testing, and validating in isolation.

DL-01 -- Mock environment construction and scenario validation Built 12 SKUs, 6 suppliers, 15 invoices, 4 injectable scenarios, FastAPI server, and live dashboard. Validated all 4 scenarios produce reproducible, resettable, non-bleeding results. Evidence: inventory.json · suppliers.json · stockout_warning.json · duplicate_invoice.json · state.py

DL-02 -- Architecture conflict resolution test Combined stockout + duplicate invoice scenario run against both Pipeline and Orchestra to confirm the sequential pattern fails and Orchestra's precedence table resolves it correctly. Evidence: graph.py · decide.py · llm.py · api_client.py

DL-04 -- Live integration and model selection Wired betsy.html to the live FastAPI endpoints, added POST /api/run-agent, switched from Mistral 7B to Llama 3.1 8B, ran stockout_warning end-to-end. Evidence: dashboard/betsy.html · server/main.py · shared/llm.py · pipeline/nodes/decide.py

DL-05 -- HITL approval queue: 11 test cases Built /api/approvals and ran 11 test cases covering: empty queue, populate on run, approve, reject, double-resolve guard (400), unknown ID guard (404). Found and fixed the state.reset() bug that cleared approvals between runs. Evidence: server/routers/approvals.py · pipeline/nodes/act.py · shared/api_client.py · server/state.py · docs/test-report-dl05.md

DL-06 -- Persistence restart test and EMA formula verification Restart test: 7 log entries and 1 pending approval reloaded from betsy.db after hard server kill -- zero entries lost. EMA formula verified to four decimal places against two known inputs. Evidence: server/db.py · server/scheduler_instance.py · server/routers/stats.py · server/routers/orders.py · tests/test_ema_learning.py · docs/test-report-dl06.md

DL-07 -- Long-term learning integration test Two real LLM pipeline runs with 8 structured delivery rounds between them. QuickShip 0.92 -> 0.44 (5 × 8-day-late), FastParts 0.95 -> 0.97 (3 × on-time). Composite crossover confirmed at round 5 with exact predicted values. Evidence: tests/test_long_term_learning.py · server/routers/suppliers.py · pipeline/nodes/evaluate.py

DL-08 -- Notification system: 22 unit tests Desktop (plyer) and email (smtplib) channels tested with mocked sends. All toggle, truncation, exception-swallowing, and config coercion cases covered. 22/22 passing in 0.27s. Evidence: server/notifier.py · server/config.py · server/routers/notifications.py · tests/test_notifier.py

Field

Testing in a realistic or real-world context -- observing actual behaviour rather than a controlled scenario.

DL-04 -- First real end-to-end agent run Ran the stockout_warning scenario against the live environment for the first time. Observed the financial safeguard trigger (\$11,937 PO > \$5,000 limit -> pending_human_review). Confirmed the safeguard is working as designed, not a failure. Evidence: images/dl04-pipeline-first-run.png ·

images/dl04-betsy-live-data.png · images/dl04-run-betsy-controls.png

Workshop

Hands-on design sessions: schema design, tradeoff structuring, environment modelling.

DL-01 -- Procurement data schema design Designed the SKU and supplier data schemas from scratch -- determining which fields (reorder_point, daily_usage_avg, lead_days, unit_price) are required for agent decision-making vs. nice-to-have. Designed real supplier tradeoffs (fast + expensive vs. cheap + slow) to ensure non-trivial decisions. Evidence: inventory.json · suppliers.json

By DL

DL	Methods used	Key evidence files
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DL-01	Lab · Workshop	inventory.json · suppliers.json · stockout_warning.json · duplicate_invoic
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DL-02	Library · Showroom · Lab	graph.py · decide.py · inventory_monitor.py · supplier_scout.py
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DL-03	Library · Showroom	docs/user_requirements.md · docs/wireframes.md · dashboard/wireframe.h
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DL-04	Lab · Field	dashboard/betsy.html · shared/llm.py · pipeline/nodes/decide.py · images/dl04
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DL-05	Lab	server/routers/approvals.py · pipeline/nodes/act.py · server/state.py · docs/test-rep
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DL-06	Lab	server/db.py · tests/test_ema_learning.py · docs/test-report-dl06.md
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DL-07	Lab	tests/test_long_term_learning.py · pipeline/nodes/evaluate.py
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DL-08	Lab	server/notifier.py · tests/test_notifier.py · server/config.py
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